Worksheet 4.6

Writing formulae for ionic compounds

Answers

Fill in the blanks in the list of electrovalencies below. Then write the formula for 10 different ionic compounds in the space provided.

+1



H^+ hydrogen



K^+ potassium



Ag^+ silver



Na^+ sodium



NH_4^+ ammonium

+2



Ca^{2+} calcium



Cu^{2+} copper (II)



Fe^{2+} iron (II)



Pb^{2+} lead

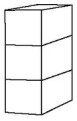


Mg^{2+} magnesium

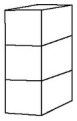


Zn^{2+} zinc

+3



Al^{3+} aluminium



Fe^{3+} iron (III)

–1



Br^- bromide



Cl^- chloride



HCO_3^- bicarbonate



OH^- hydroxide



I^- iodide



NO_3^- nitrate

–2



CO_3^{2-} carbonate



O^{2-} oxide

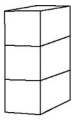


SO_4^{2-} sulfate

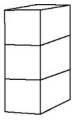


S^{2-} sulfide

–3



PO_4^{3-} phosphate



N^{3-} nitride

Examples include: KBr (potassium bromide), AgCl (silver chloride), $NaHCO_3$ (sodium bicarbonate), NH_4I (ammonium iodide), Ag_2O (silver oxide), Na_2SO_4 (sodium sulfate), $Cu_3(PO_4)_2$ (copper phosphate)

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